

PAPER_1167_UPDATE_Master_Lagrangian_6_Term

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PAPER_1167_UPDATE — Master Lagrangian 6-Term Form

Original: PAPER_1167 UQFF Master Lagrangian

Update: Session June 2026

Status: Update

Session Extensions

The closed 6-term form is wired:

$$\mathcal{L}_{\rm UQFF} = \frac{R_{26}}{2\kappa_E} - \frac{1}{4}F^2_{\rm DPM} + \sum_i \beta_i U_g U_b + |U_m|^2 + \frac{1}{2}|\partial U_A|^2 + \frac{25}{12}\rho_{\rm SCm}\left[(U_A/v_{UA})^2 - 1\right]^2$$

β_i ladder $\beta_i = \frac{3(5-n)}{20}$ text{ with } $n \in \{1,2,3,4\}$ yielding $\beta_1 = 0.6$, $\beta_2 = 0.45$, $\beta_3 = 0.30$, $\beta_4 = 0.15$.

For $n \approx 0.98067$ (fractional permitted): $\beta = 0.6029$ canonical.

Variational stationarity $\delta S/\delta\phi = 0$ enforces $F_U = 1$ globally.